College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

ASSIGNMENT 02 — WRITTEN QUESTION PAPER

Lessons Covered: (01) Hello World, JDK & Java Program Anatomy | (02) Variables, Data Types & Console I/O

Duration: 1 Hour **Total Marks:** 40 **Answer all questions** in your own answer booklet / sheets.

Instructions to Candidates

1. This is a **written paper**. Answer **all** questions in the answer sheets provided by you.
2. Do **not** write answers on this question paper.
3. Write clearly using blue or black pen. Number your answers to match the question numbers.
4. Where Java code is required, write syntactically correct code as taught in class (VS Code / JDK).
5. Marks for each question are shown in brackets. Manage your time carefully.
6. Mobile phones and programmable devices are not allowed.

Part A — Short Answers (JDK, Tools & Program Anatomy) [12 Marks]

Question 1

a. What does **JDK** stand for? State one reason why the JDK is required to write and run Java programs. **(2 marks)**

b. Name the **IDE** that students are allowed to use in this module, and name the Microsoft extension pack required for Java. **(2 marks)**

c. Write the two terminal commands used to *compile* and then *run* a file named `Hello.java` (class name `Hello`). **(2 marks)**

Question 2

Study the incomplete Java program below.

```
________ class Welcome
{
    public static void ________(String[] args)
    {
        System.out.________("Hello, World!");
    }
}
```

a. Fill in the three blanks so the program is a valid Hello World program. **(3 marks)**

b. What is the purpose of the `main` method in a Java application? **(1 mark)**

c. State the difference between `//` and `;` in Java. **(2 marks)**

Part B — Variables, Data Types & Console I/O [14 Marks]

Question 3

For each situation below, state the most suitable Java data type and write one example **declaration** and **initialization** as two steps. Present your answer as a numbered list (1–5) or a neat table in your answer sheet. **(5 marks)**

1. A student's age in whole years
2. Course fee with cents (e.g. 15000.50)
3. Student's full name
4. Whether the student has paid (yes/no)
5. A single letter grade (e.g. A)

Question 4

a. Which quotes are used for a `String` value, and which for a `char` value? Give one example of each. **(2 marks)**

b. Write the syntax of creating an object of the `Calculator` class. **(2 marks)**

c. Write the two code lines that create an object in Scanner Class and reads a line from terminal in scanner object. **(3 marks)**

d. Write one Java statement that prints: *Hello Kasun, you are 20* using variables `name` and `age` with string concatenation. **(2 marks)**

Part C — Code Writing & Tracing **[14 Marks]**

Question 5 — Trace the output

What is printed by the following program? Show the final console output clearly. **(4 marks)**

```
public class Trace01 {
    public static void main(String[] args) {
        int qty = 3;
        double price = 25.50;
        String item = "Pen";
        double total = qty * price;
        System.out.println(item + " x " + qty);
        System.out.println("Total = " + total);
    }
}
```

Question 6 — Find and correct errors

The program below has **four** errors. For each error, state what is wrong and write the corrected line. **(4 marks)**

```
public class Broken
{
    public static void Main(String args)
    {
        string name = 'Nimal';
        System.out.println("Hi " + name)
    }
}
```

Question 7 — Write a complete program

Write a complete Java program named `StudentCard` that: **(6 marks)**

1. Creates a `Scanner` and reads the student's **name** (text) and **mark** (whole number) from the keyboard.
2. Stores them in suitable variables.
3. Prints a line in the form: `Student: <name> | Mark: <mark>`

Include the class, `main` method, and any required `import`. A package declaration is optional.

******* END OF QUESTION PAPER *******
